

Consumer Theory I: Choices, Preferences and Utility Functions

1 Big Picture

1. The consumer (or any decision-maker) has a **consumption set** (or choice set), $\mathbf{X} = \mathbb{R}_+^n$, containing all conceivable alternatives.
2. The consumer's circumstances (e.g. budget constraint) limit the alternatives he or she is actually able to achieve, that is, the **feasible set**, $B \subset \mathbf{X}$.
3. His or her inclinations or attitudes toward the choices are described by the **preference relation** $\succsim$ defined on $\mathbb{R}_+^n$.
4. If the consumer's preference relation $\succsim$ is **complete**, **transitive**, **continuous**, **strictly monotonic**, and **strictly convex** on $\mathbb{R}_+^n$, it can be represented by a real-valued **utility function**, u , that is **continuous**, **strictly increasing** and **strictly quasiconcave** on $\mathbb{R}_+^n$.
5. The consumer is motivated to choose the most preferred feasible alternative according to his or her preference relation. Formally,

$$\mathbf{x}^* \in B \text{ such that } \mathbf{x}^* \succsim \mathbf{x} \text{ for all } \mathbf{x} \in B$$

6. Or the consumer's problem

$$\max_{\mathbf{x} \in \mathbb{R}_+^n} u(\mathbf{x}) \quad s.t. \quad \mathbf{p}\mathbf{x} \leq \mathbf{I}$$

2 Preference

1. **Consumption set**, $\mathbf{X} = \mathbb{R}_+^n$: the set of all alternatives, or complete **consumption plans**.

2. **Consumption plan**, $\mathbf{x} = (x_1, \dots, x_n)$, be a vector containing different quantities of each of the n commodities, $\mathbf{x} \in \mathbf{X}$.

3. Properties of consumption set $\mathbf{X}$

- Nonempty, $\emptyset \neq \mathbf{X}$
- $\mathbf{X}$ is closed

$\mathbf{X}$ is a closed set if every convergent sequence in $\mathbf{X}$ converges to a point in $\mathbf{X}$.

- $\mathbf{X}$ is convex

$$\mathbf{x}, \mathbf{y} \in \mathbf{X} \implies \forall t \in [0, 1] \quad t\mathbf{x} + (1 - t)\mathbf{y} \in \mathbf{X}.$$

- $\mathbf{0} \in \mathbf{X}$.

2.1 Preference relation

Preferences: tells us how consumer would rank any two consumption plans (bundles), *assuming the plans were available at no cost*.

- If consumer i is willing to give up the bundle $\mathbf{y}$ in return for the bundle $\mathbf{x}$, she has, in effect, ranked bundle $\mathbf{x}$ higher than $\mathbf{y}$. We express such a preference ranking by writing $\mathbf{x} \succsim \mathbf{y}$ (Consumer i (weakly) prefers $\mathbf{x}$ to $\mathbf{y}$, or $\mathbf{x}$ is at least as good as $\mathbf{y}$.)
- If the consumer is also willing to give up $\mathbf{x}$ in order to get bundle $\mathbf{y}$ back again so that $\mathbf{y} \succsim \mathbf{x}$, then the two bundles are equally ranked and we write $\mathbf{x} \sim \mathbf{y}$ (Consumer i is indifferent between $\mathbf{x}$ and $\mathbf{y}$).
- If the consumer is not willing to freely switch back to bundle $\mathbf{y}$ ($\mathbf{y} \not\succsim \mathbf{x}$) then $\mathbf{x}$ is ranked strictly higher than $\mathbf{y}$ and we write $\mathbf{x} \succ \mathbf{y}$. (Consumer i strictly prefers $\mathbf{x}$ to $\mathbf{y}$).

2.1.1 Axioms of consumer choices

- **Completeness:** For any two bundles $\mathbf{x}$ and $\mathbf{x}'$, it is that either $\mathbf{x} \succsim \mathbf{x}'$ or $\mathbf{x}' \succsim \mathbf{x}$.

or Completeness: For any two bundles $\mathbf{x}$ and $\mathbf{x}'$, it is that either $\mathbf{x} \succ \mathbf{x}'$, $\mathbf{x}' \succ \mathbf{x}$, or $\mathbf{x} \sim \mathbf{x}'$.

- **Transitivity:** For any three choices $\mathbf{x}$, $\mathbf{x}'$, and $\mathbf{x}''$,

$$\text{if } \mathbf{x} \succsim \mathbf{x}' \text{ and } \mathbf{x}' \succsim \mathbf{x}'', \text{ then } \mathbf{x} \succsim \mathbf{x}''.$$

or Transitivity: For any three choices $\mathbf{x}$, $\mathbf{x}'$, and $\mathbf{x}''$,

$$\text{if } \mathbf{x} \succ \mathbf{x}' \text{ and } \mathbf{x}' \succ \mathbf{x}'', \text{ then } \mathbf{x} \succ \mathbf{x}'',$$

and

$$\text{if } \mathbf{x} \sim \mathbf{x}' \text{ and } \mathbf{x}' \sim \mathbf{x}'', \text{ then } \mathbf{x} \sim \mathbf{x}'.$$

- **Continuity.** For any consumption bundle $\mathbf{x}$, the set of consumption bundles that is weakly preferred to $\mathbf{x}$ $\{\mathbf{y} : \mathbf{y} \succeq \mathbf{x}\}$, and the no better than set $\{\mathbf{y} : \mathbf{y} \preceq \mathbf{x}\}$ are closed. Another definition: For any sequence of pairs $\{(\mathbf{w}^n, \mathbf{z}^n)\}_{n=1}^{\infty}$ with $\mathbf{w}^n \succeq \mathbf{z}^n$ for all n , $\mathbf{w} = \lim_{n \rightarrow \infty} \mathbf{w}^n$, and $\mathbf{z} = \lim_{n \rightarrow \infty} \mathbf{z}^n$, we have $\mathbf{w} \succeq \mathbf{z}$. (See MWG.)
- **Local Nonsatiation:** The preference $\succeq$ is locally nonsatiated if for all $\mathbf{x} \in \mathbf{X}$ and for all $\varepsilon > 0$, there exists $y \in \mathbf{X}$ such that $\|y - \mathbf{x}\| \leq \varepsilon$ and $y \succ \mathbf{x}$.
- **Strict monotonicity:** The preference $\succeq$ is strictly monotone if

$$\forall \mathbf{x}, y \in \mathbf{X}, y \geq \mathbf{x} \implies y \succ \mathbf{x}$$

while

$$y \gg \mathbf{x} \implies y \succ \mathbf{x}.$$

- **Convexity.** If $\mathbf{x} \succeq \mathbf{x}'$, then for all $t \in [0, 1]$, $t\mathbf{x} + (1 - t)\mathbf{x}' \succeq \mathbf{x}'$.

Strict convexity. If $\mathbf{x} \neq \mathbf{x}'$ and $\mathbf{x} \succeq \mathbf{x}'$, then for all $t \in (0, 1)$, $t\mathbf{x} + (1 - t)\mathbf{x}' \succ \mathbf{x}'$.

3 Utility function

1. A utility function defined over $\mathbf{X}$ assigns a real number to each member of $\mathbf{X}$. We say a utility function $u : \mathbf{R}_+^n \rightarrow \mathbf{R}$ represents a preference relation $\succeq$ if for any objects $\mathbf{x}$ and $\mathbf{x}'$,

$$\mathbf{x} \succeq \mathbf{x}' \text{ iff } u(\mathbf{x}) \geq u(\mathbf{x}').$$

2. Utility function is a convenient way to describe a preference relation. For example, if I tell you that my preference over apples and bananas is $u(a, b) = a^{0.5} + b$, then you would know how I would choose between any combinations of apples and bananas.
3. Theorem (Jehle & Reny Theorem 1.1): If a preference relation $\succeq$ is complete, transitive, continuous, and strictly monotonic, then there exists a continuous $u : \mathbf{R}_+^n \rightarrow \mathbf{R}$ that represents $\succeq$.

4. Step 1. Let $e \equiv (1, 1, \dots, 1)$ denote a unit consumption bundle (which contains one unit of each good). For any consumption bundle $\mathbf{x}$, define $u(\mathbf{x})$ by the condition

$$u(\mathbf{x})e \sim \mathbf{x}.$$

($u(\mathbf{x})e \equiv (u(\mathbf{x}), \dots, u(\mathbf{x}))$, i.e., $u(\mathbf{x})e$ is the consumption bundle that consists of $u(\mathbf{x})$ units of each good.) Step 2. $u(\mathbf{x})$ exists and is uniquely defined for all $\mathbf{x}$. By the continuity of preferences, the sets

$$A \equiv \{t \in \mathbb{R}_+ : te \succsim \mathbf{x}\}$$

and

$$B \equiv \{t \in \mathbb{R}_+ : \mathbf{x} \succsim te\}$$

are both closed. By strictly monotonicity, for all $t' \geq t$, $t \in A$ implies $t' \in A$. Hence

$$A = \{t \in [\underline{t}, \infty)\}$$

and

$$B = \{t \in [0, \bar{t}]\}.$$

By completeness, any t must belong to either A or B ; thus $A \cup B = \mathbb{R}_+$. We therefore can conclude that $\underline{t} \leq \bar{t}$, or that the intersection of A and B is non-empty. Let $t^* \in A \cap B$, then $t^*e \sim \mathbf{x}$, and set $u(\mathbf{x}) = t^*$. Uniqueness follows from the strictly monotonicity axiom. If $t_1e \sim \mathbf{x}$ and $t_2e \sim \mathbf{x}$, then by the transitivity, $t_1e \sim t_2e$. By strictly monotonicity, it must be $t_1 = t_2 = t^*$. Step 3. For any two consumption bundles $\mathbf{x}$ and $\mathbf{x}'$,

$$\begin{aligned} \mathbf{x} \succsim \mathbf{x}' &\Leftrightarrow u(\mathbf{x})e \succsim u(\mathbf{x}')e \quad (\text{transitivity}) \\ &\Leftrightarrow u(\mathbf{x}) \geq u(\mathbf{x}') \quad (\text{strict monotonicity}). \end{aligned}$$

Hence, $\mathbf{x} \succsim \mathbf{x}'$ if and only if $u(\mathbf{x}) \geq u(\mathbf{x}')$.

Step

4. It remains only to show that u is continuous. To prove continuity, we need to make use of the following mathematical result: A function is continuous if for any open set $(a, b) \subset \mathbb{R}$, the inverse mapping $u^{-1}((a, b))$ is open in $\mathbb{R}_+^n$ (Theorem A1.6). In our case,

$$\begin{aligned} u^{-1}((a, b)) &= \{\mathbf{x} | a < u(\mathbf{x}) < b\} \\ &= \{\mathbf{x} | ae \prec u(\mathbf{x})e \prec be\} \\ &= \{\mathbf{x} | ae \prec \mathbf{x} \prec be\}. \end{aligned}$$

Finally, the intersection of two open set is open, or $u^{-1}((a, b))$ is open.

3.1 Property of the utility function u

1. The consumer's preference relation $\succsim$ which has those properties can be represented by a real-valued **utility function**, u , that is **continuous**, **strictly increasing** and **strictly quasiconcave** on $\mathbb{R}_+^n$.
2. **Continuity.** A function u is continuous at $\mathbf{x}_0$ iff for all $\varepsilon > 0$, there exists some δ such that for all $\mathbf{x}' \in \{\mathbf{x} : d(\mathbf{x}, \mathbf{x}_0) \leq \delta\}$, $|u(\mathbf{x}) - u(\mathbf{x}_0)| \leq \varepsilon$. Intuitively, if u is continuous at $\mathbf{x}$, a small movement from $\mathbf{x}$ will produce only a small change in u .

3. Concavity and Convexity

- A function $u(\mathbf{x}) : \mathbb{R}^n \rightarrow \mathbb{R}$ is concave iff for all $\mathbf{x}, y \in \mathbb{R}^n$ and for all $t \in [0, 1]$,

$$tu(\mathbf{x}) + (1 - t)u(y) \leq u(t\mathbf{x} + (1 - t)y).$$

- A function $u(\mathbf{x}) : \mathbb{R}^n \rightarrow \mathbb{R}$ is convex iff for all $\mathbf{x}, y \in \mathbb{R}^n$ and for all $t \in [0, 1]$,

$$tu(\mathbf{x}) + (1 - t)u(y) \geq u(t\mathbf{x} + (1 - t)y).$$

4. Quasiconcavity and Quasiconvexity

- A function is quasiconcave if for any $\mathbf{x}, \mathbf{x}'$, and $t \in [0, 1]$

$$u(t\mathbf{x} + (1 - t)\mathbf{x}') \geq \min[u(\mathbf{x}), u(\mathbf{x}')].$$

A function is strictly quasiconcave if for any $\mathbf{x}, \mathbf{x}'$, and $t \in (0, 1)$

$$u(t\mathbf{x} + (1 - t)\mathbf{x}') > \min[u(\mathbf{x}), u(\mathbf{x}')].$$

- A function is quasiconvex if for any $\mathbf{x}, \mathbf{x}'$, and $t \in [0, 1]$

$$u(t\mathbf{x} + (1 - t)\mathbf{x}') \leq \max[u(\mathbf{x}), u(\mathbf{x}')].$$

A function is strictly quasiconvex if for any $\mathbf{x}, \mathbf{x}'$, and $t \in (0, 1)$

$$u(t\mathbf{x} + (1 - t)\mathbf{x}') < \max[u(\mathbf{x}), u(\mathbf{x}')].$$

5. Concave $\Rightarrow$ quasiconcave; convex $\Rightarrow$ quasiconvex

6. Note that $\succsim$ is convex does not imply that u is also convex. The term “convexity” have different meanings depending whether it is referred to a preference relation, a function, or a set.

7. (JR, Theorem 1.3). $u(\mathbf{x})$ is (strictly) quasiconcave if and only if $\succsim$ is (strictly) convex. $u(x)$ is strictly increasing if and only if $\succsim$ is strictly monotonic. The claim follows directly from the definition.
8. Recall from intermediate micro that the marginal rate of substitution between good 1 and good 2 is the amount of good 2 a consumer is willing to give up for one extra unit of good 1. Mathematically:

$$MRS \equiv -\frac{dx_2}{dx_1}|_{U=U^*}.$$

If MRS equals 2, then you are willing to pay 2 units of x_2 for 1 unit of x_1 . Mathematically, the indifference curve is represented by a function $x_2(x_1)$ such that $U(x_1, x_2(x_1)) = U^*$, where U^* is the level of utility. Thus

$$\begin{aligned} \frac{\partial U}{\partial x_1} + \frac{\partial U}{\partial x_2} \frac{dx_2}{dx_1} &= 0 \\ \Leftrightarrow -\frac{dx_2}{dx_1} &= \frac{\frac{\partial U}{\partial x_1}}{\frac{\partial U}{\partial x_2}} \end{aligned}$$

A standard assumption is that $dMRS/dx_1 < 0$, that is, MRS is decreasing—as the consumer consumes more of good 1, he is willing to pay for good 1 (in terms of good 2) decreases. Another way to put it is that the consumer prefers balanced consumption bundles (ie. some of everything).

9. Convex preferences is the extension of DMRS (diminishing marginal rate of substitution) to the n goods case.

3.1.1 More on Concavity and Convexity

- Suppose u is continuously differentiable. Then u is concave if and only if, for any $\mathbf{x}, y$,

$$u(\mathbf{x}) + \nabla u(\mathbf{x})(y - \mathbf{x}) \geq u(y)$$

where

$$\nabla u(\mathbf{x}) \equiv \left(\frac{\partial u(\mathbf{x})}{\partial \mathbf{x}_1}, \dots, \frac{\partial u(\mathbf{x})}{\partial \mathbf{x}_n} \right)$$

so

$$\nabla u(\mathbf{x})(y - \mathbf{x}) = \sum_{i=1}^n \frac{\partial u(\mathbf{x})}{\partial \mathbf{x}_i} (y_i - x_i).$$

- If the utility function is twice continuously differentiable, we can check the concavity of $u(\cdot)$ by using the Hessian matrix

$$H = \begin{pmatrix} \frac{\partial^2 u}{\partial x_1^2} & \frac{\partial^2 u}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 u}{\partial x_1 \partial x_n} \\ \frac{\partial^2 u}{\partial x_2 \partial x_1} & \frac{\partial^2 u}{\partial x_2^2} & \cdots & \frac{\partial^2 u}{\partial x_2 \partial x_n} \\ \cdots & \cdots & \cdots & \cdots \\ \frac{\partial^2 u}{\partial x_n \partial x_1} & \frac{\partial^2 u}{\partial x_n \partial x_2} & \cdots & \frac{\partial^2 u}{\partial x_n^2} \end{pmatrix}$$

The leading principal minors $|H|$

$$|H_1| = \frac{\partial^2 u}{\partial x_1^2}, |H_2| = \begin{vmatrix} \frac{\partial^2 u}{\partial x_1^2} & \frac{\partial^2 u}{\partial x_1 \partial x_2} \\ \frac{\partial^2 u}{\partial x_2 \partial x_1} & \frac{\partial^2 u}{\partial x_2^2} \end{vmatrix} \dots |H_n| = |H|$$

The sufficient condition for u to be concave is $|H_1| \leq 0, |H_2| \geq 0, \dots$; or $|H_n| \leq 0$ if n is odd, $|H_n| \geq 0$ if n is even. That is, the Hessian matrix is negative semidefinite.

If $|H_1| < 0, |H_2| > 0, \dots$; or $|H_n| < 0$ if n is odd, $|H_n| > 0$ if n is even, or the Hessian matrix is negative definite, u is strictly concave.

- Suppose u is continuously differentiable, u is convex if and only if, for any $\mathbf{x}, y$,

$$u(\mathbf{x}) + \nabla u(\mathbf{x})(y - \mathbf{x}) \leq u(y)$$

- If the utility function is twice continuously differentiable, the sufficient condition for u to be convex is $|H_1| \geq 0, |H_2| \geq 0, \dots, |H_n| \geq 0$, or the Hessian matrix is positive semidefinite. If $|H_1| > 0, |H_2| > 0, \dots, |H_n| > 0$ or the Hessian matrix is positive definite, u is strictly convex.
- Suppose u is continuously differentiable. Then u is quasiconcave if and only if $u(y) \geq u(\mathbf{x})$ implies that

$$\nabla u(\mathbf{x})(y - \mathbf{x}) \geq 0,$$

This condition means that the the better-than set lies on upper-right side of the tangent to u at $\mathbf{x}$.

- If the utility function is twice continuously differentiable, we can check the quasiconcavity of $u(\cdot)$ by using the bordered Hessian matrix

$$\bar{H} = \begin{pmatrix} 0 & \frac{\partial u}{\partial x_1} & \frac{\partial u}{\partial x_2} & \cdots & \frac{\partial u}{\partial x_n} \\ \frac{\partial u}{\partial x_1} & \frac{\partial^2 u}{\partial x_1^2} & \frac{\partial^2 u}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 u}{\partial x_1 \partial x_n} \\ \frac{\partial u}{\partial x_2} & \frac{\partial^2 u}{\partial x_2 \partial x_1} & \frac{\partial^2 u}{\partial x_2^2} & \cdots & \frac{\partial^2 u}{\partial x_2 \partial x_n} \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ \frac{\partial u}{\partial x_n} & \frac{\partial^2 u}{\partial x_n \partial x_1} & \frac{\partial^2 u}{\partial x_n \partial x_2} & \cdots & \frac{\partial^2 u}{\partial x_n^2} \end{pmatrix}$$

If $u(\cdot)$ is quasiconcave, we have $|\bar{H}_2| \leq 0, |\bar{H}_3| \geq 0, \dots$

If for all $\mathbf{x} \geq \mathbf{0}$, we have $|\bar{H}_2| < 0, |\bar{H}_3| > 0, \dots$ then $u(\mathbf{x})$ is quasiconcave on the nonnegative orthant $\mathbb{R}_+^n$.

If for all $\mathbf{x} \gg \mathbf{0}$, we have $|\overline{H}_2| < 0, |\overline{H}_3| > 0, \dots$ then $u(\mathbf{x})$ is strictly quasiconcave on the (strictly) positive orthant.

3.2 Ordinality v.s. Cardinality

1. A monotonic transformation is a function f that is strictly increasing. Examples: adding a constant, take square root.
2. (JR, Theorem 1.2). Many utility functions represent the same underlying preferences. In fact, if u represents a preference relation $\succsim$ then any monotonic transformation of u , denote $v(x) = f(u(x))$, will also represent the same preference relation.
3. The only information conveyed by a utility function is about whether a bundle has a higher utility than another bundle. It does not tell you by how much you like the first bundle over the second.
4. We call a measure that conveys information about ranking an ordinal measure. By contrast, a cardinal measure measures intensity as well as ranking.

3.3 Some Common Preferences and Utility Functions

1. The curvature of the indifference curves or the magnitude of the first derivative of MRS tells us how fast consumers become satiated with the good. Here are some commonly used utility functions:
2. Linear utility function:

$$u(\mathbf{x}) = \sum_{i=1}^n \alpha_i x_i$$

with $\alpha_i \geq 0$. Let $t = \max_i \alpha_i / p_i$. Demand function:

$$x_i = \begin{cases} 0 & \text{if } \frac{\alpha_i}{p_i} < t \\ c_i & \text{if } \frac{\alpha_i}{p_i} = t \end{cases}$$

with $c_i \geq 0$ and $\sum_{i \in \{j | \alpha_j / p_j = t\}} p_j c_j = y$.

3. Leontief utility function:

$$u(\mathbf{x}) = \min(\alpha_i x_i)$$

with $\alpha_i > 0$. Demand function:

$$x_i = \frac{y}{\alpha_i \sum_{j=1}^n p_j / \alpha_j}.$$

4. Cobb-Douglas utility function:

$$u(\mathbf{x}) = \prod_{i=1}^n x_i^{\alpha_i},$$

with $\alpha_i > 0$ and $\sum_{i=1}^n \alpha_i = 1$. Demand:

$$\mathbf{x}_i(p, y) = \frac{\alpha_i y}{p_i}.$$

5. Constant-elasticity-of-substitution utility function (CES):

$$u(\mathbf{x}) = \left(\sum_{i=1}^n \alpha_i x_i^\rho \right)^{1/\rho}$$

with $\alpha_i > 0$. Note that this function becomes linear when $\rho = 1$, Cobb-Douglas when $\rho = 0$, and Leontief when $\rho = -\infty$. Demand function:

$$x_i = \frac{p_i^{\frac{1}{\rho-1}} \alpha_i^{\frac{-1}{\rho-1}}}{\sum_{j=1}^n p_j^{\frac{\rho}{\rho-1}} \alpha_j^{\frac{-1}{\rho-1}}} y.$$

Definition: Elasticity of Substitution between good i and good j :

$$-\frac{\partial (\mathbf{x}_j / \mathbf{x}_i)}{\partial (p_i / p_j)} \frac{p_i / p_j}{\mathbf{x}_j / \mathbf{x}_i}.$$

The elasticity of substitution for CES function is $1/(1 - \rho)$.

6. Stone-Geary utility function has the form

$$u(\mathbf{x}) = \prod_{i=1}^n (\mathbf{x}_i - a_i)^{b_i},$$

where $a_i \geq 0$, $b_i \geq 0$ and $\sum_{i=1}^n b_i = 1$. This function only makes sense when $y > \sum_{i=1}^n p_i a_i$. Demand function:

$$\mathbf{x}_i = \frac{b_i (y - \sum p_i a_i)}{p_i} + a_i.$$

7. Homothetic Preferences.

- (a) Homogeneous functions. A function is said to be homogeneous of degree r , if multiplication of each of the independent variables by a constant t will alter the value of the function by the proportion t^r , i.e.,

$$u(t\mathbf{x}_1, \dots, t\mathbf{x}_n) = t^r u(\mathbf{x}_1, \dots, \mathbf{x}_n).$$

- (b) Homothetic functions. A homothetic function is a composite function in the form

$$u = U(f(\mathbf{x}_1, \dots, \mathbf{x}_n)), \quad U'(f) \neq 0$$

where $f(\dots)$ is homogeneous of degree r .

In economics we often impose the restriction that $U'(f) > 0$, that is, u is a monotonic transformation of the homogeneous function $f(\cdot)$.

- (c) Homothetic preference implies a linear expansion path: the optimal consumption bundles at different income level, assuming fixed prices, lie on a straight line emanating from the origin.

4 References

1. JR: Chapter 1; Chapter A1;
2. MWG: Chapters 1 & 2; Mathematical Appendix
3. AR: Lectures 1 & 2.